Project Submission: Student Notes Application

1. GitHub Link

The complete source code, container configurations, and orchestration manifests for this project are hosted on GitHub.

- **Repository URL:** https://github.com/TechTAnish-07/student-notes-app

This repository includes the backend logic, Dockerfile, and all necessary Kubernetes YAML files (app-deployment, app-service, db-deployment, and db-service-secret).

2. Project Overview

This project is a robust, two-tier student notes application designed to demonstrate modern containerization and orchestration best practices. The goal was to build a fully functional, production-ready system that showcases proficiency across the full development and deployment lifecycle.

- **Application Purpose and Features:** The application serves as an efficient digital tool for students to manage their academic data. It supports full CRUD (Create, Read, Update, Delete) operations, allowing users to:
    - **Create** new notes, capturing essential metadata like a specific *Subject* and *Title*, alongside the detailed *Content*.
    - **View** all stored notes or retrieve specific entries for review.
    - **Delete** outdated or irrelevant information to maintain a clean workspace.
- The simple, API-driven design ensures a smooth, focused user experience for academic note management.

- **Architecture and Decoupling:** The application employs a decoupled, two-tier architecture:
    - **1. Application Tier:** Handled by the Node.js/Express backend, responsible for all business logic, routing, and API interactions.
    - **2. Data Tier:** Managed by the PostgreSQL database, ensuring data integrity and persistence.
- This separation enhances scalability, as the application tier can be scaled independently of the database, and improves maintainability by clearly separating concerns.
- **Technology Stack and Deployment:** The project leverages powerful, industry-standard technologies for reliability and portability:
    - **Backend:** Node.js and Express provide a lightweight, high-performance environment suitable for handling asynchronous I/O and serving RESTful APIs.
    - **Database:** PostgreSQL was selected for its robust relational data model, ensuring reliable storage and complex query capabilities for structured note data.
    - **Containerization (Docker):** The entire application environment (including the database setup) is encapsulated in Docker containers, guaranteeing consistency and eliminating "works on my machine" issues across different environments.
    - **Orchestration (Kubernetes):** Kubernetes is used for automated deployment, scaling, and management. It provides critical features like self-healing, rolling updates, and secure management of sensitive data (database credentials) via Kubernetes Secrets. This demonstrates readiness for enterprise-level deployment and traffic management.

3. Step-by-Step Instructions

Follow these terminal commands to build, deploy, and verify the application.

Build Docker Image

To package the application into a container image, run the following command from the root directory:

```
docker build -t notes-app .
```

Kubernetes Deployment

Apply the configuration files in the specific order listed below to ensure dependencies (like secrets and databases) are established before the application starts:

1. **Apply Secrets:** `kubectl apply -f k8s/db-service-secret.yaml`
2. **Deploy Database:** `kubectl apply -f k8s/db-deployment.yaml`
3. **Deploy Application:** `kubectl apply -f k8s/app-deployment.yaml`
4. **Expose Service:** `kubectl apply -f k8s/app-service.yaml`

Check Status and Access

Verify that all components are running correctly using these commands:

- **Check Pods:** `kubectl get pods`
- **Check Services:** `kubectl get services`

If you are running in a local environment without a LoadBalancer, use port forwarding to access the app:

```
kubectl port-forward svc/notes-app 3000:3000
```

Summary of Component Roles

File Name	Responsibility
db-service-secret.yaml	Manages sensitive database credentials.

File Name	Responsibility
db-deployment.yaml	Manages the PostgreSQL database pod and persistent volume.
app-deployment.yaml	Manages the Node.js application pods and environment variables.
app-service.yaml	Provides a stable network endpoint for the application.

Output

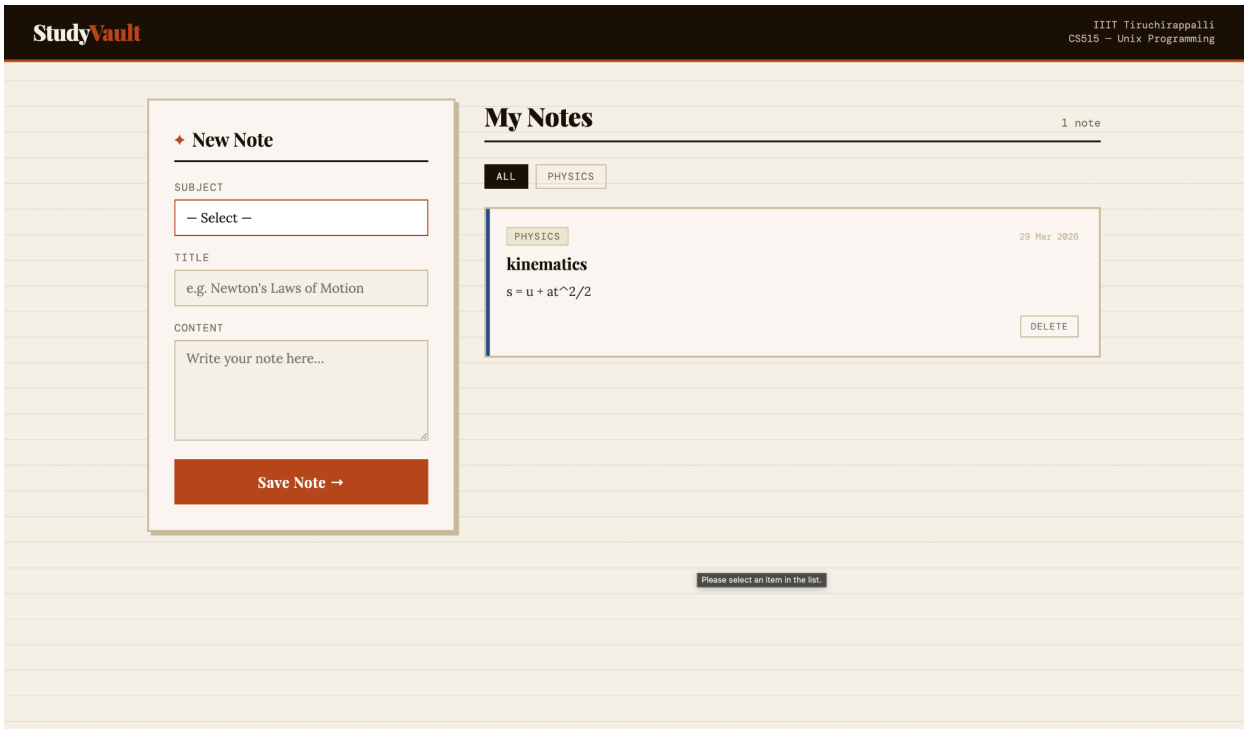
Created by Tanish patidar

Date: Mar 28, 2026

```
tanish@TAnishs-MacBook-Air assignment1 % cd files
tanish@TAnishs-MacBook-Air files % kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
notes-app-566dbb779c-gw95v    1/1     Running   0          21m
notes-app-566dbb779c-rbqkm    1/1     Running   0          21m
notes-db-6dcb494865-zvxnc     1/1     Running   0          21m
tanish@TAnishs-MacBook-Air files %
```

,

```
tanish@TAnishs-MacBook-Air files % git log
commit cfd951dcac3d57eaee969a6850a517b321c79b26 (HEAD -> main, origin/main)
Author: TechTAnish-07 <patidar29tanish@gmail.com>
Date:   Sun Mar 29 21:34:44 2026 +0530

    Added package-lock.json for reproducible builds

commit f6b7a19595133ea66856a77fa3eedaba4d3c67a3
Author: TechTAnish-07 <patidar29tanish@gmail.com>
Date:   Sun Mar 29 21:25:05 2026 +0530

    Added Kubernetes manifests: Deployment, Service, Secret, PVC

commit d61783bf39ed79ea062bfa0c35260f87b3e7d92c
Author: TechTAnish-07 <patidar29tanish@gmail.com>
Date:   Sun Mar 29 21:25:05 2026 +0530

    Added frontend UI for creating and viewing notes

commit ae52750d2e7b8d1c6e81992fc2feeeed1a82c0d8
Author: TechTAnish-07 <patidar29tanish@gmail.com>
Date:   Sun Mar 29 21:21:43 2026 +0530

    Added Dockerfile with multi-stage build and docker-compose

commit 139f54db76f74f20a2af8ae6a9433ab1e2b65459
Author: TechTAnish-07 <patidar29tanish@gmail.com>
Date:   Sun Mar 29 21:21:42 2026 +0530

    Added Express server and PostgreSQL connection logic
tanish@TAnishs-MacBook-Air files %
```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New Note
SUBJECT
— Select —
TITLE
e.g. Newton's Laws of Motion
CONTENT
Write your note here...
Save Note →
My Notes
1 note
ALL
PHYSICS
PHYSICS
29 Mar 2026
kinematics
s = u + at^2/2
DELETE
Please select an item in the list.